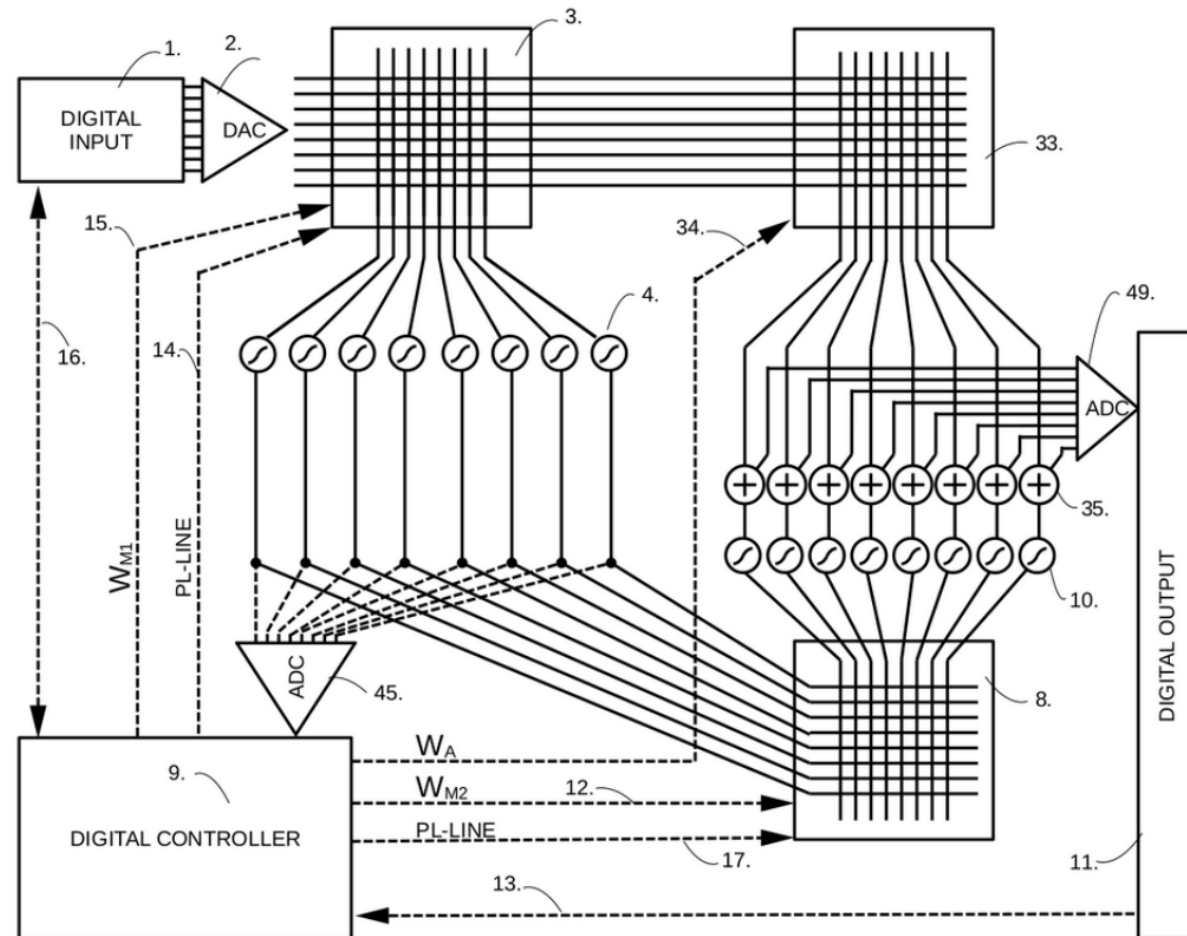
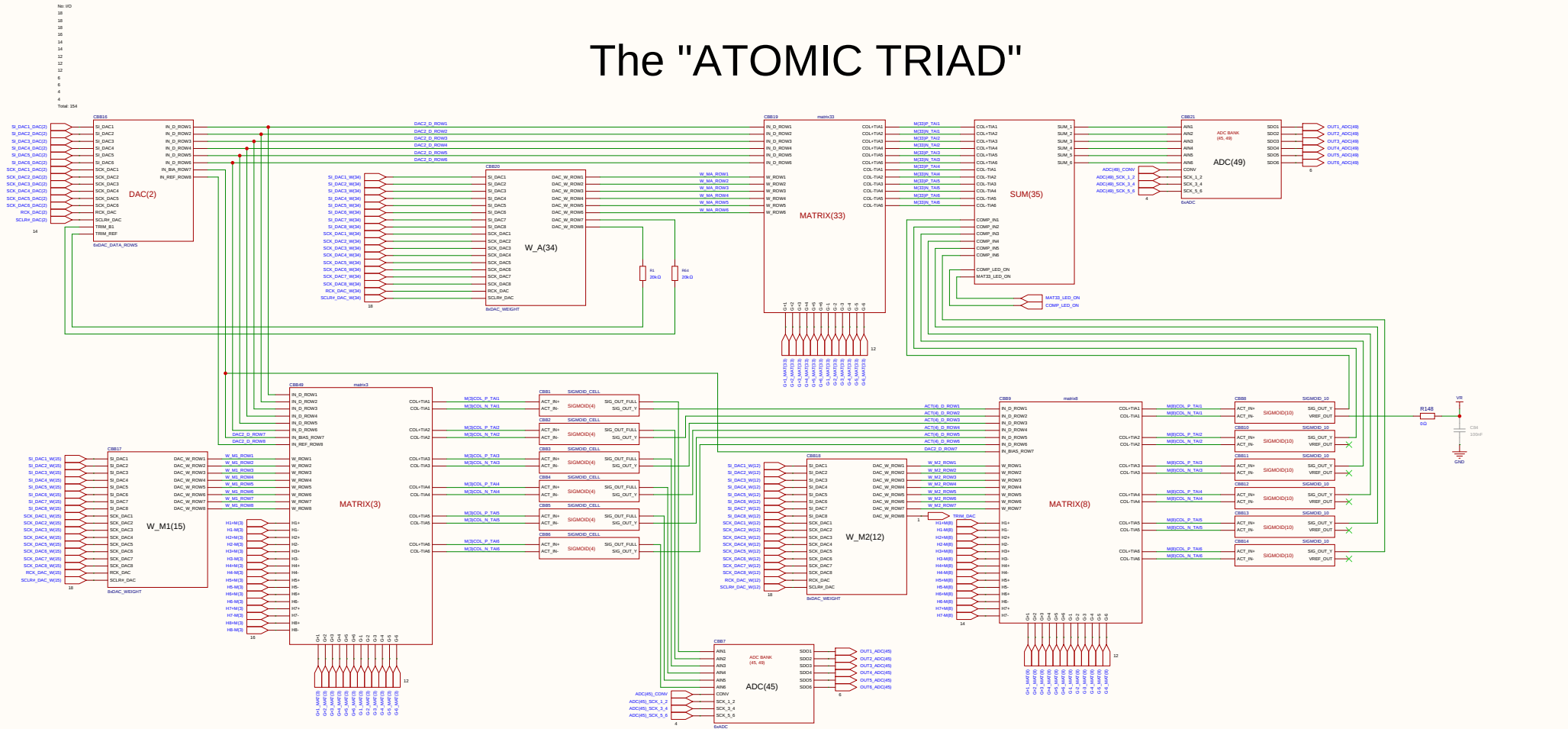


PRIORITY DATE: 2026-06-03



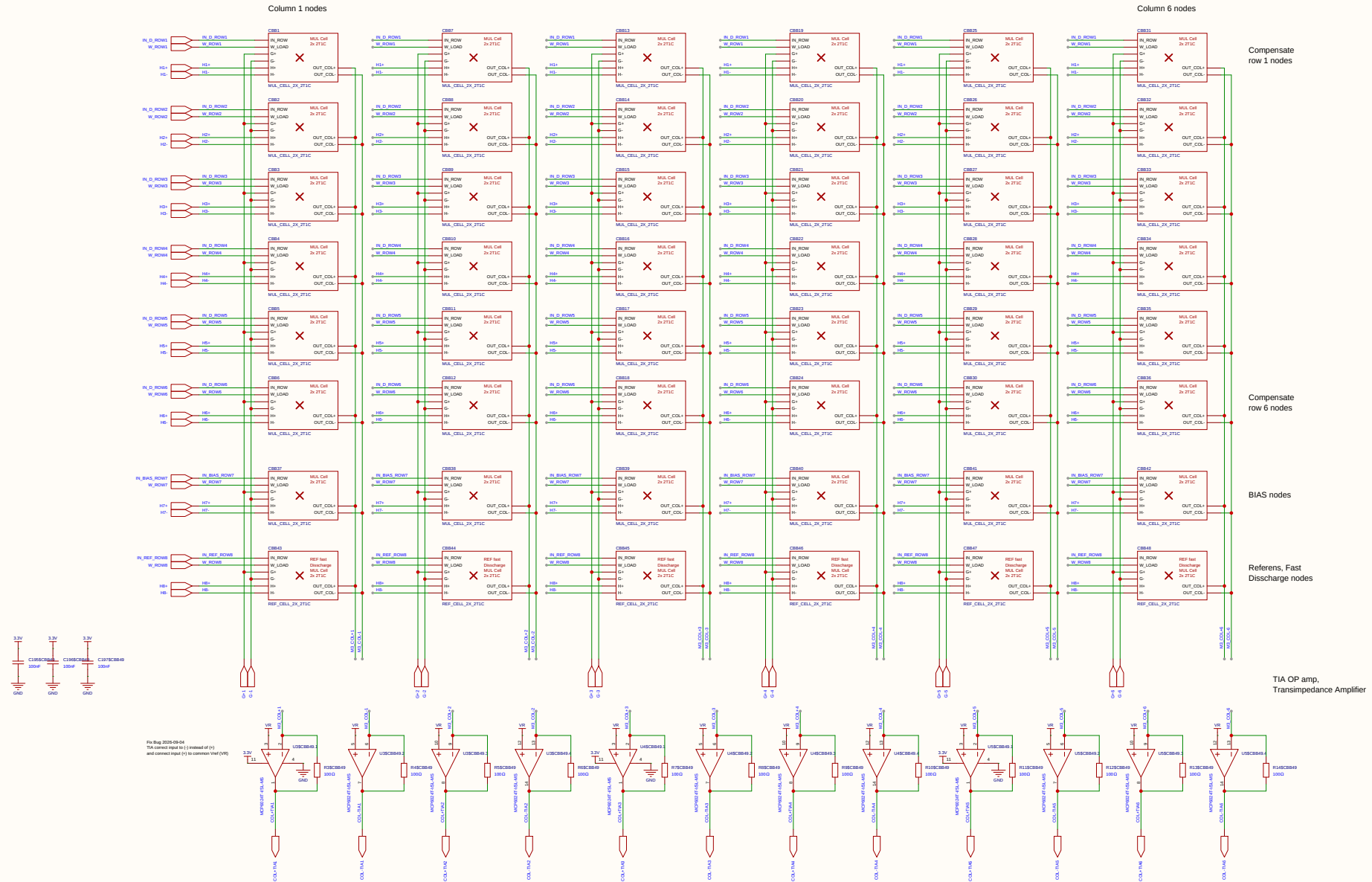
Schematic	Inverted_MAML			Create at	2026-06-19
				Update at	2026-06-19
Board	Analog_Matrix			Page	cover
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 5	
		V1.0	A4	EasyEDA.com	

The "ATOMIC TRIAD"



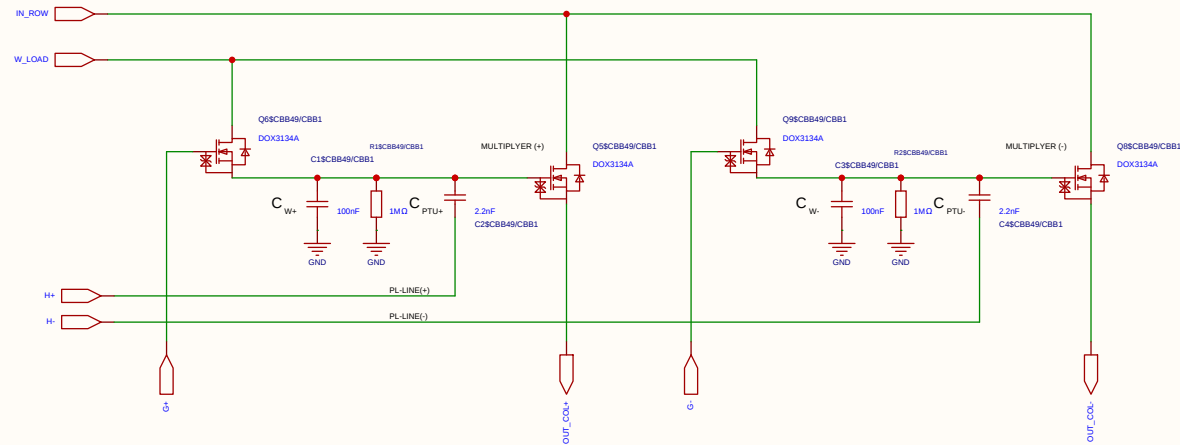
Schematic	Inverted_MAML		Create at	2026-07-19
Board	Analog_Matrix		Update at	2026-09-04
Drawn	Olle Welin	Inverted_MAML_6x6	Page	Atomic_Triad
Reviewed				
		Version	Size	Page 2 Total 5
		V1.0	A4	EasyEDA.com

$$V_{ut_TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$



Schematic	matrix3		Create at	2026-06-27
Board			Update at	2026-09-04
Drawn			Page	max3
Reviewed			Inverted_MAML_6x6	
		Version	Size	Page 1 Total 1
		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

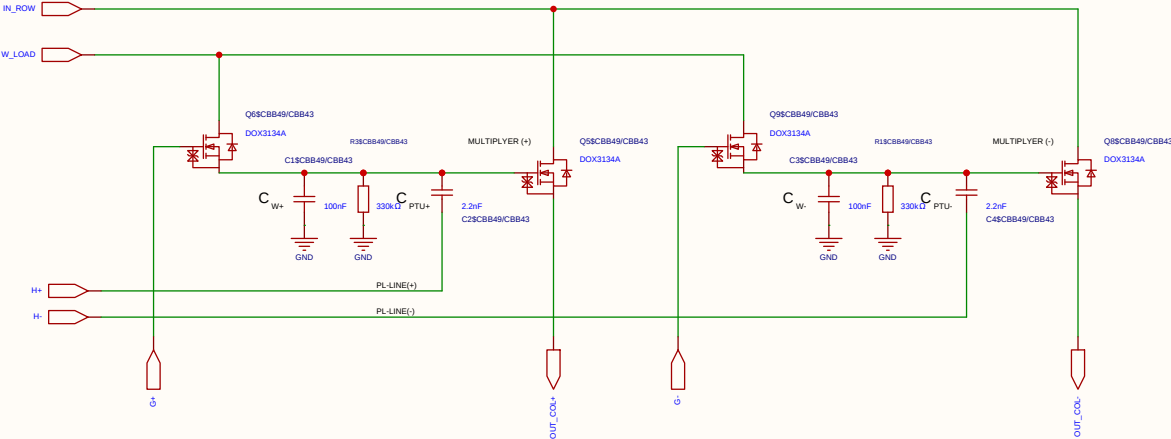
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_S) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-07-18
				Update at	2026-07-18
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)		
COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

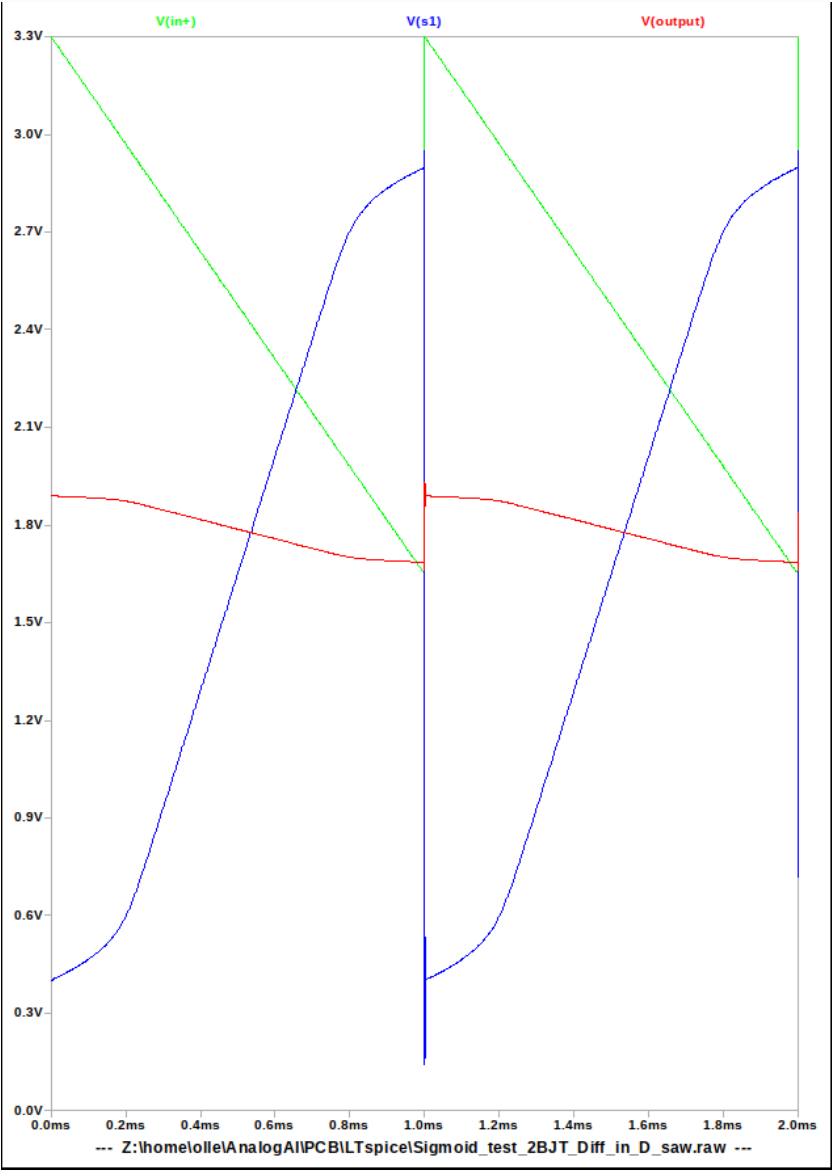
4. Voltage Levels on the PL Row			
STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

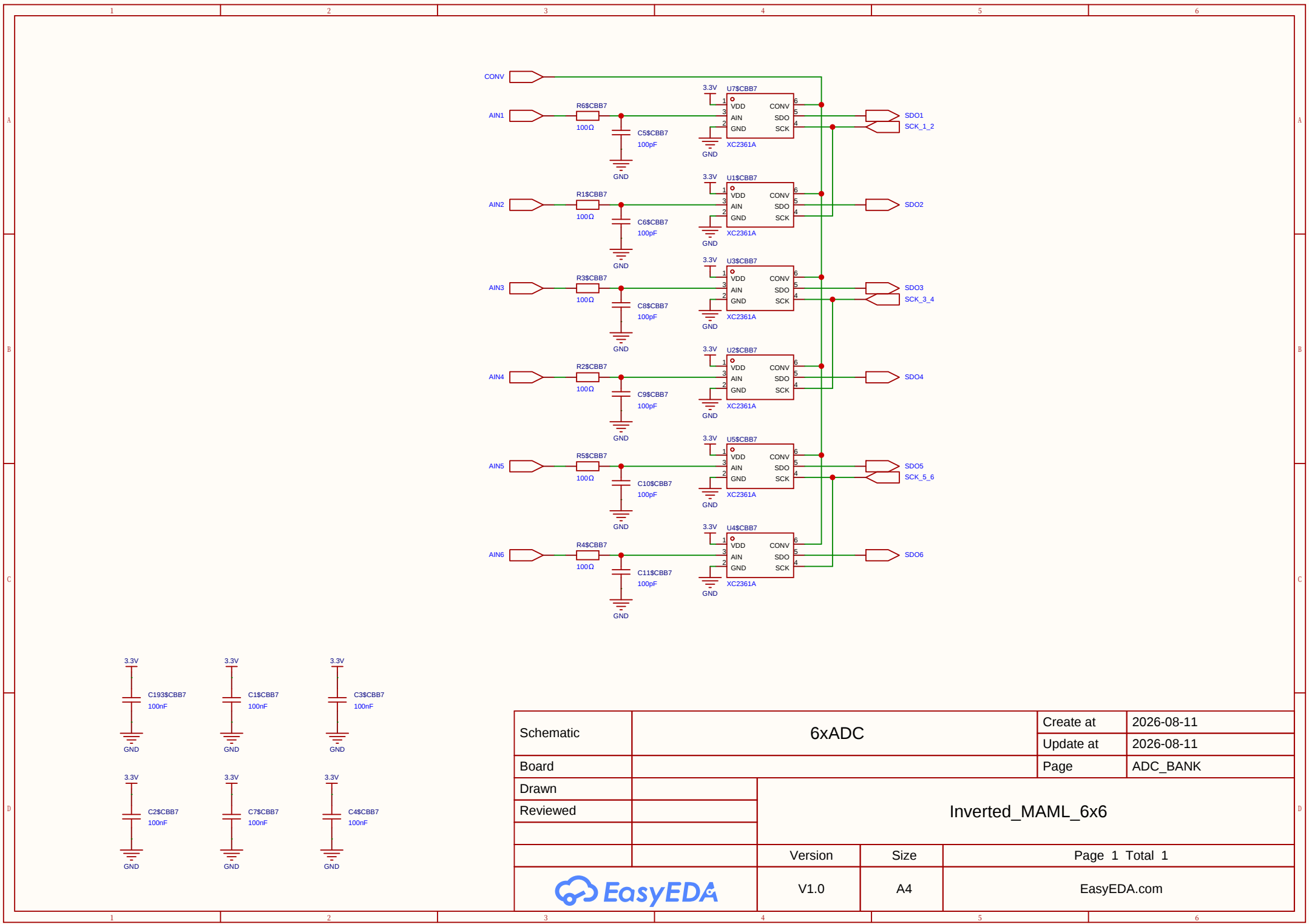
This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

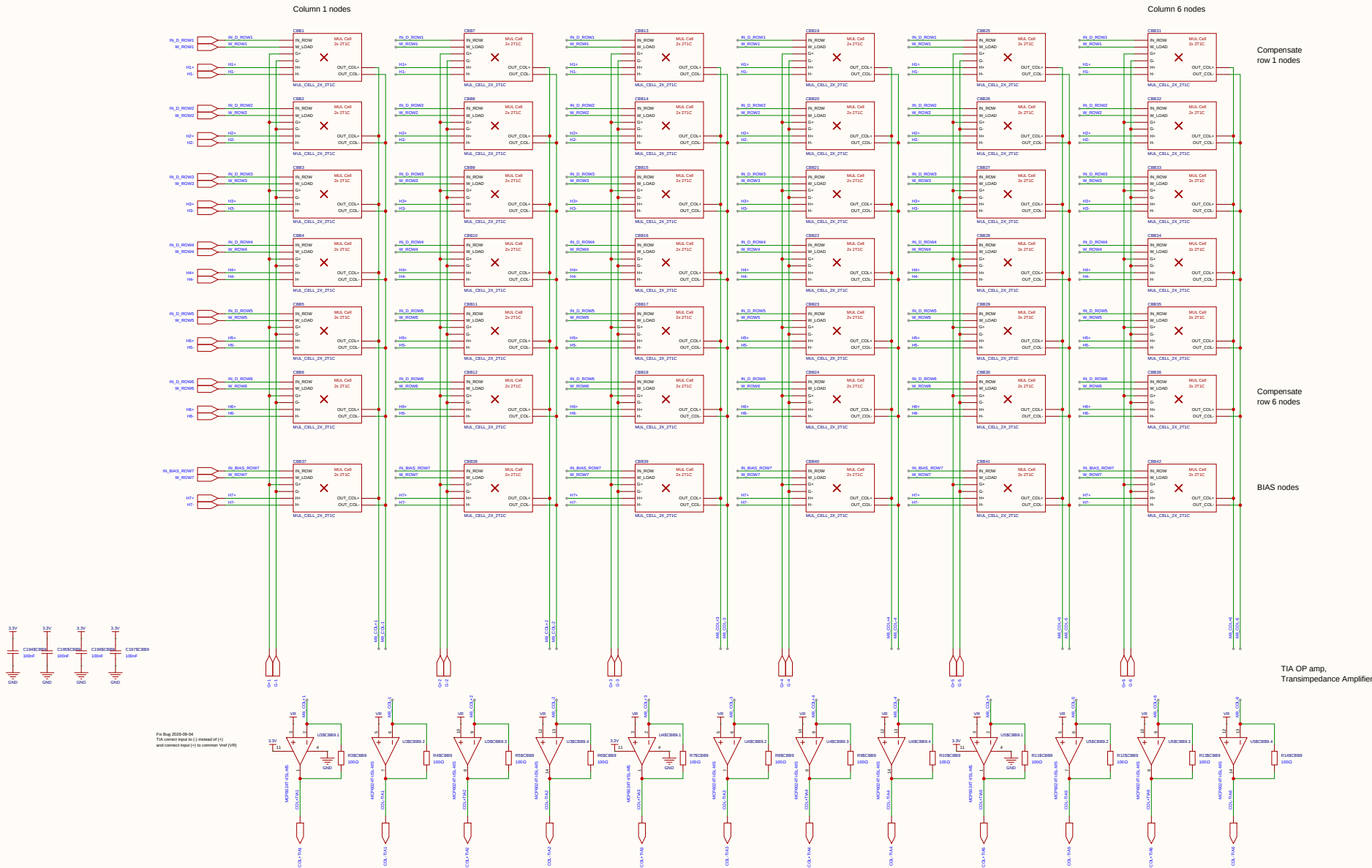
Schematic	REF_CELL_2X_2T1C		Create at	2026-07-18
			Update at	2026-07-18
Board			Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com



Schematic	SIGMOID_CELL			Create at	2026-07-07
				Update at	2026-07-07
Board				Page	SIMOID(4)
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

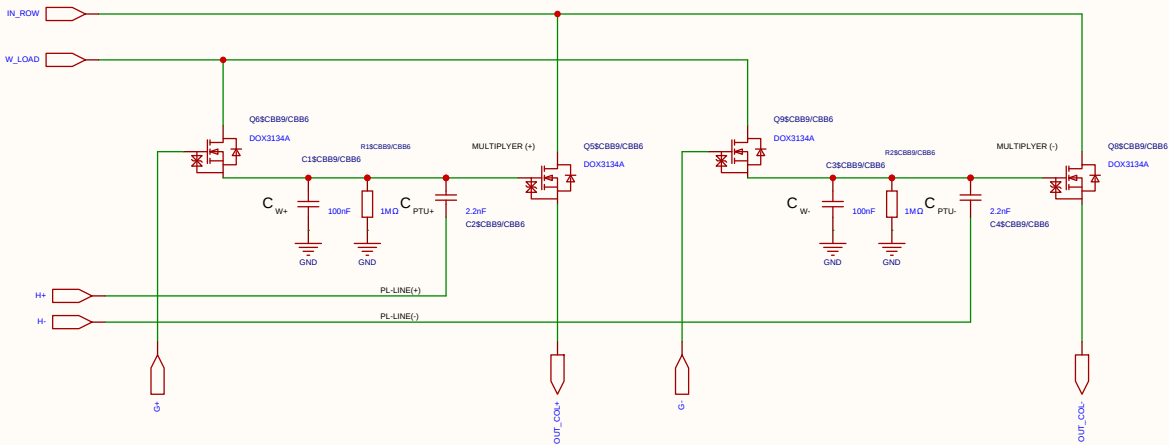


$$V_{ut_TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$



Schematic	matrix8		Create at	2026-06-27
Board			Update at	2026-09-04
Drawn			Page	max8
Reviewed			Inverted_MAML_6x6	
		Version	Size	Page 1 Total 1
		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

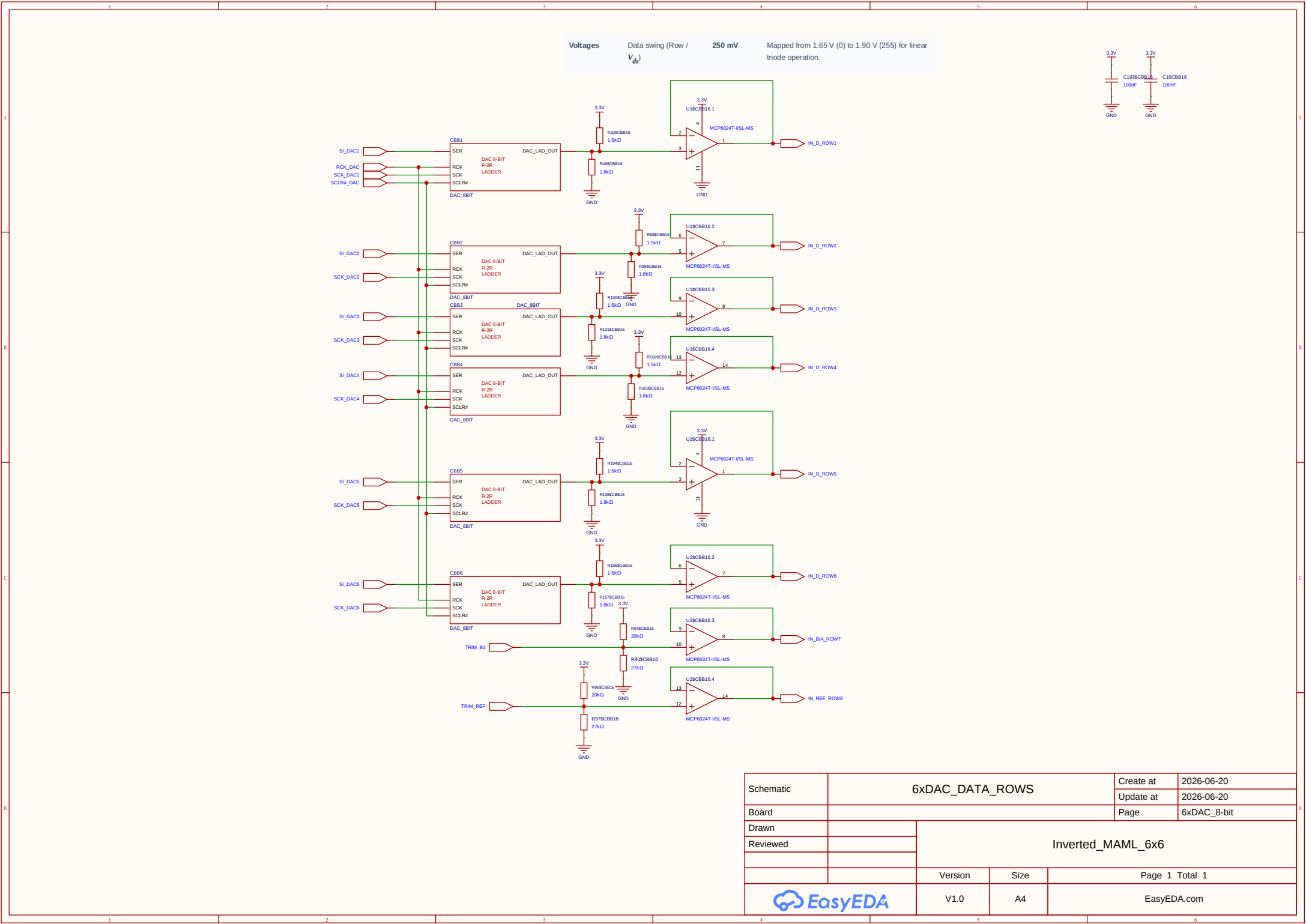
STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

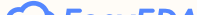
2. Resistor Network (Per PL Row)

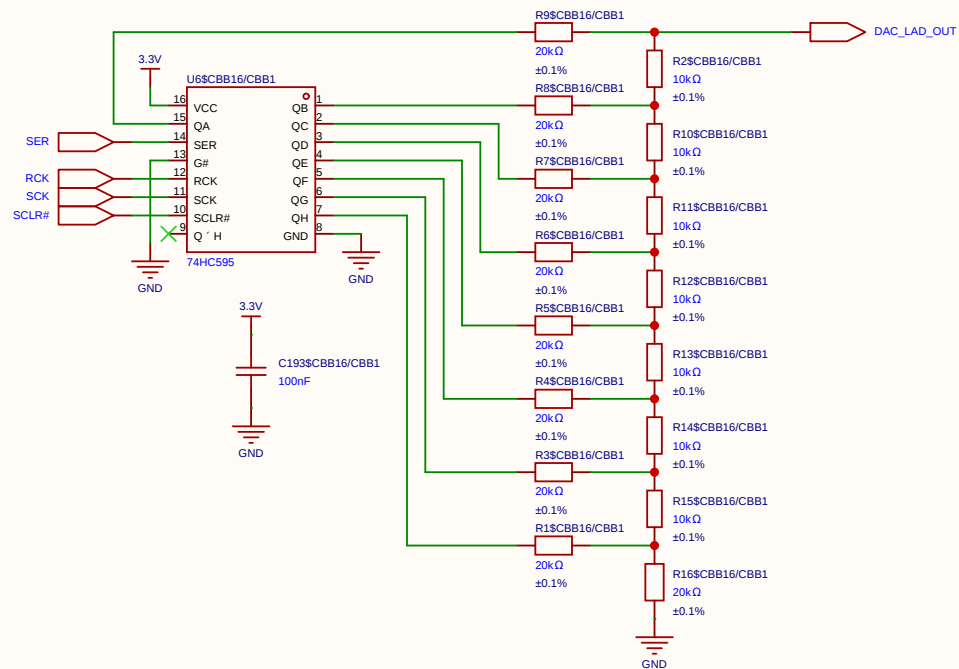
This network connects the FPGA output pin to the PL_I row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C		Create at	2026-07-18
			Update at	2026-07-18
Board			Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

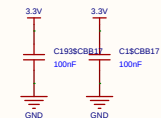
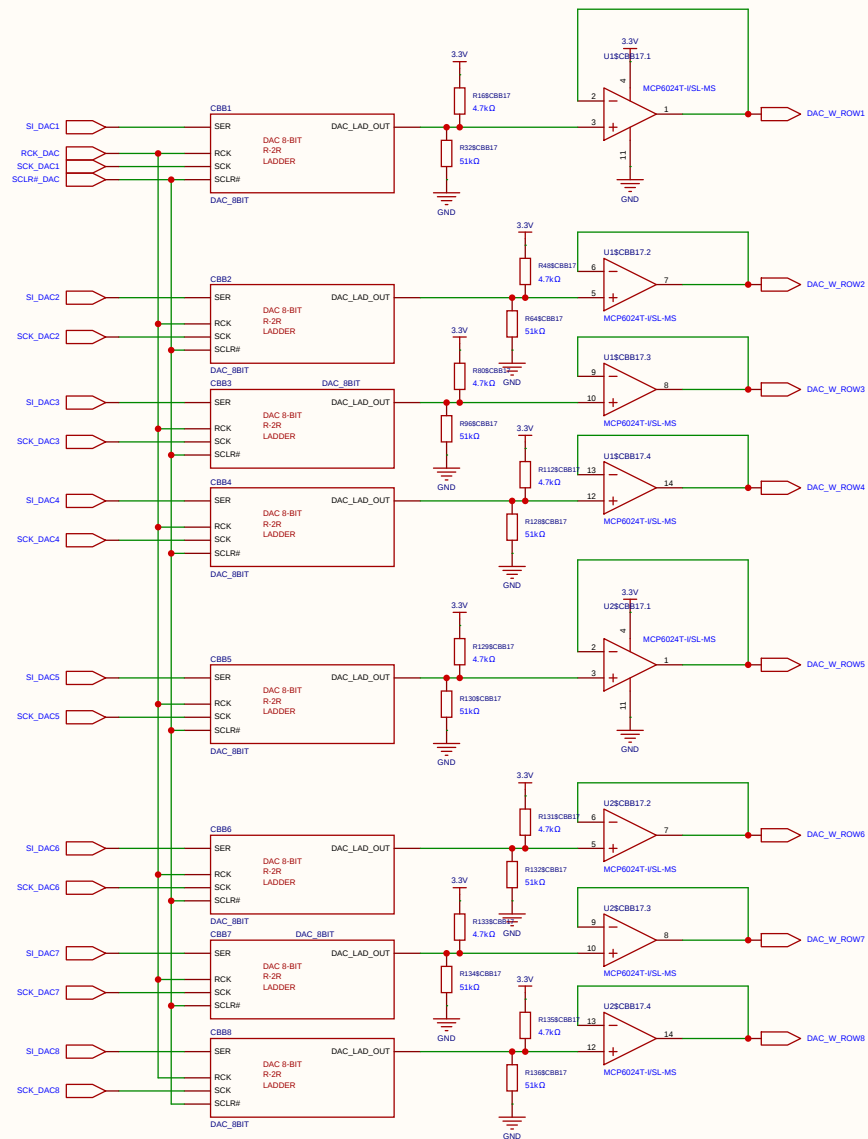


Schematic	6xDAC_DATA_ROWS			Create at	2026-06-20
				Update at	2026-06-20
Board				Page	6xDAC_8-bit
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	



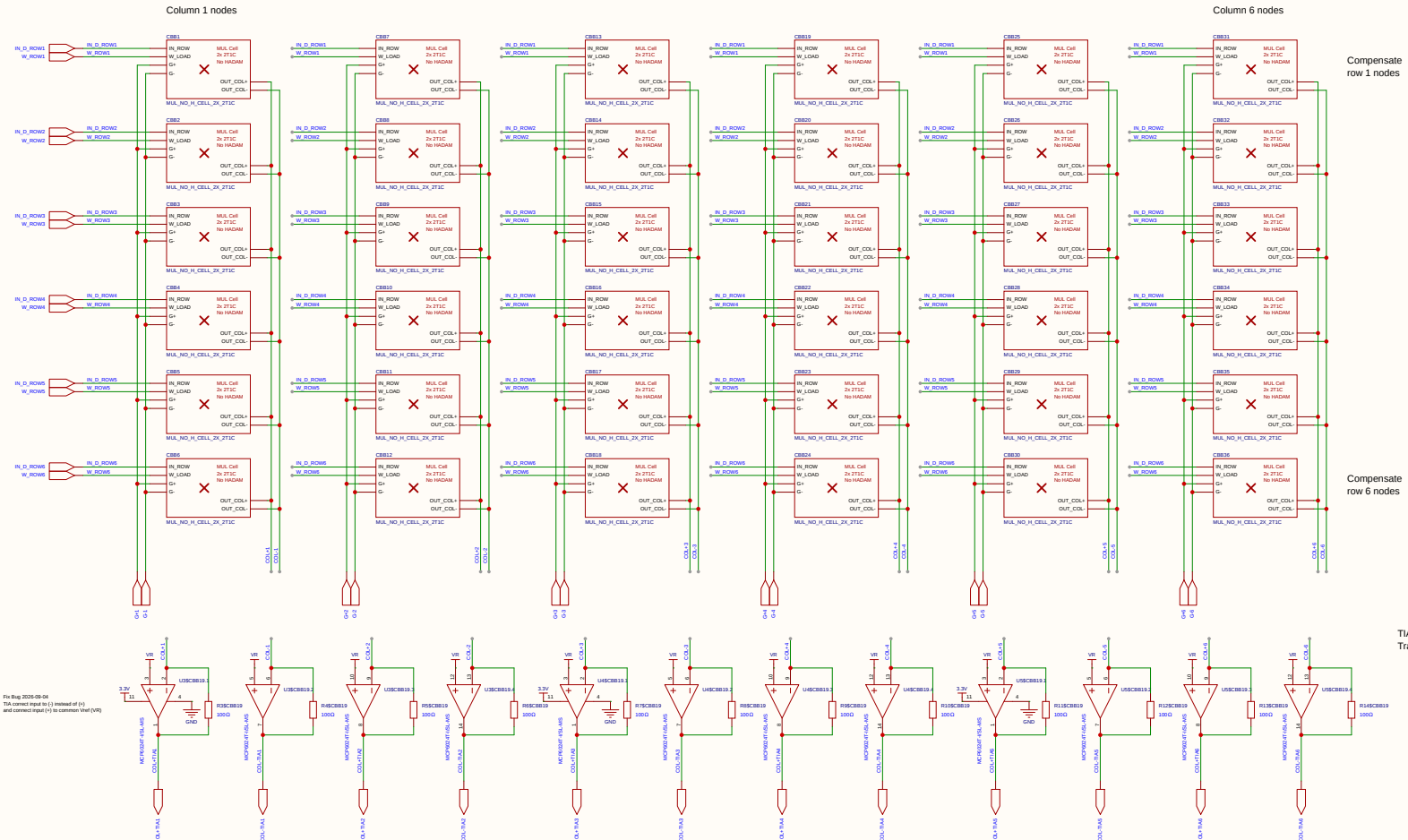
Schematic	DAC_8BIT			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	DAC_8B_R_2R
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Weight swing (Gate / V_{gs}) **1.0 V** Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).



Schematic	8xDAC_WEIGHT			Create at	2026-06-30
				Update at	2026-06-30
Board				Page	8xDAC_8-bit
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

$$V_{ut,TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$

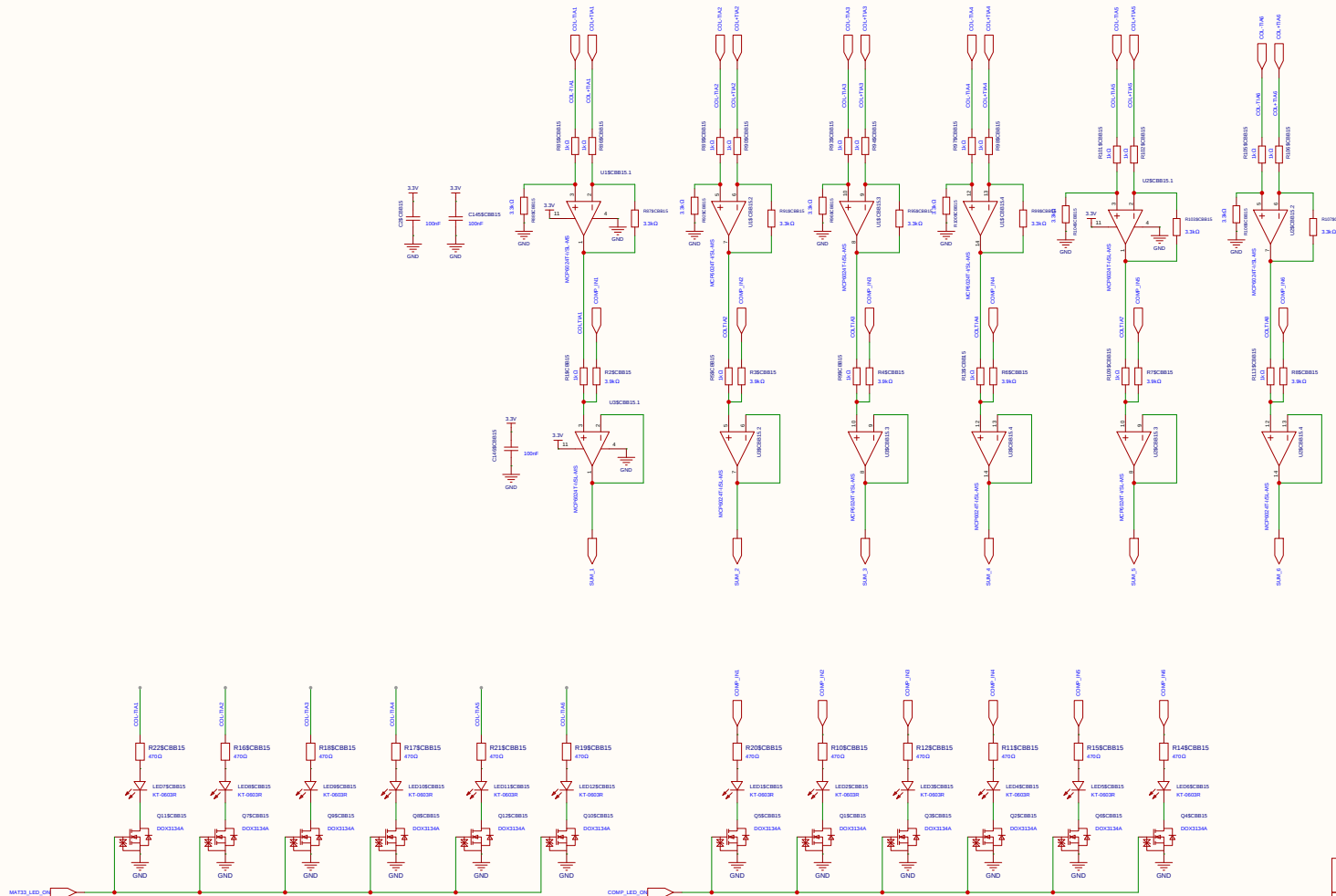


Schematic	matrix33		Create at	2026-06-28
Board			Update at	2026-09-04
Drawn	Olle Welin	Inverted_MAML_6x6	Page	max33
Reviewed				
	Version	Size	Page 1 Total 1	
	V1.0	A4	EasyEDA.com	

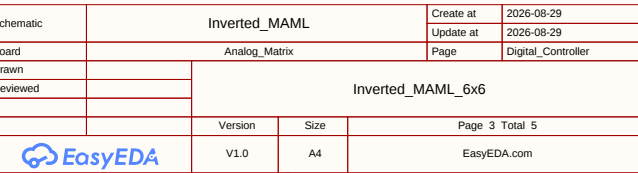
Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-28
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

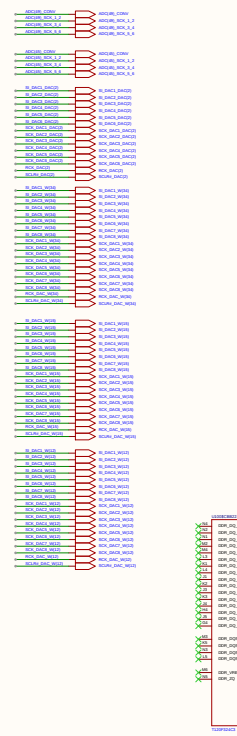
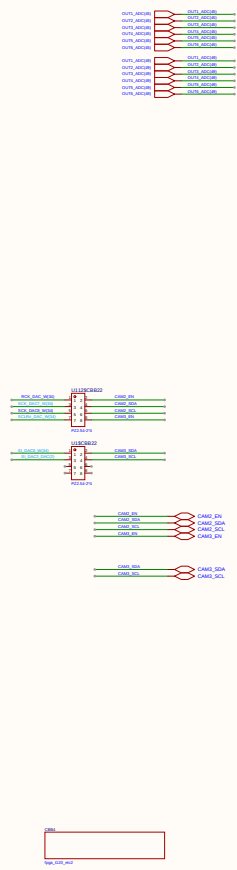
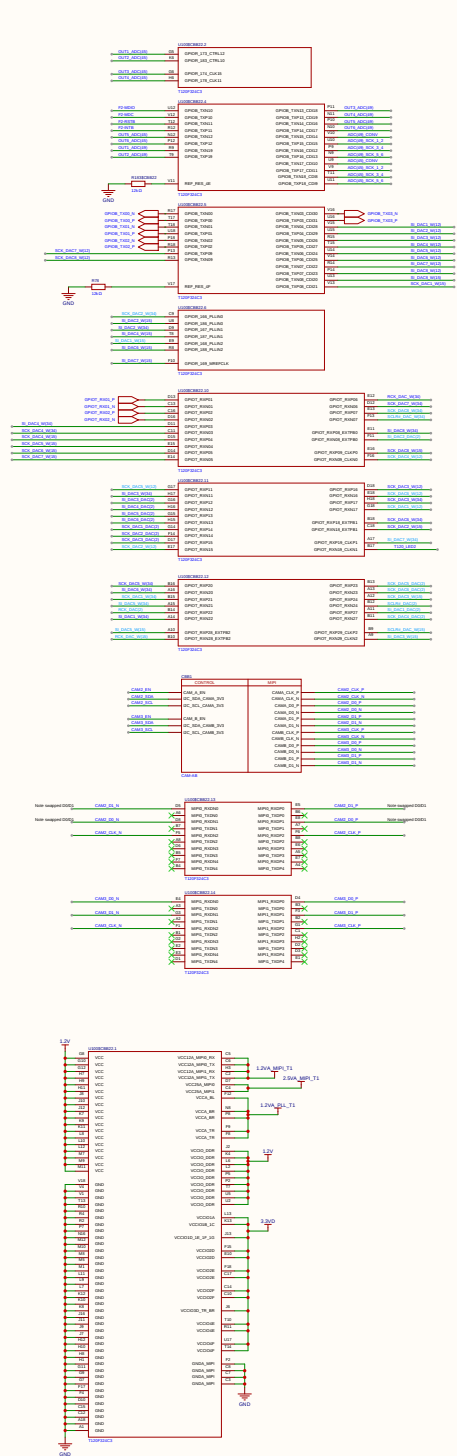
- **Calibration Example:** To match a 3.3 V ADC, the resistors are configured for a Gain of 6.6×.

$$V_{\text{ADC}} = 6.6 \times (1.65 \text{ V} - V_{\text{out_TIA}})$$



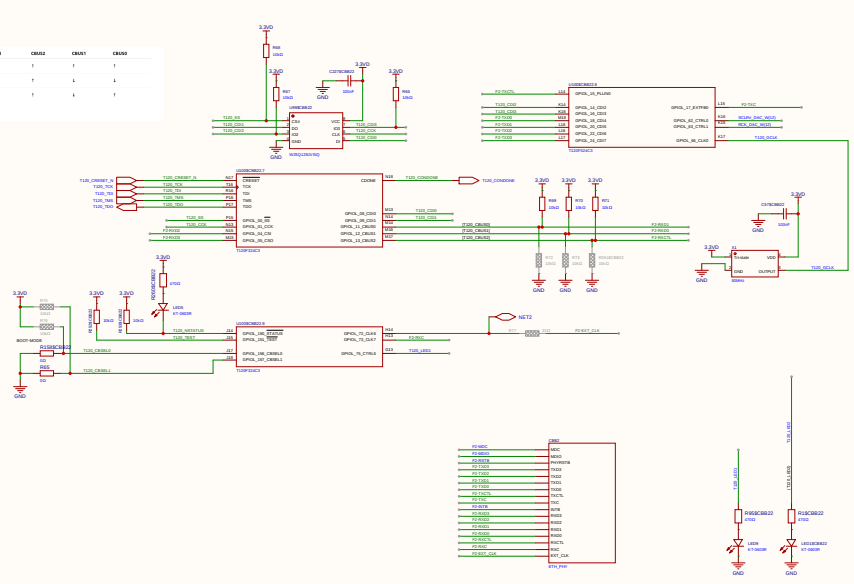
Schematic	SUM35		Create at	2026-08-06
Board			Update at	2026-08-06
Drawn	Olle Welin	Inverted_MAML_6x6	Page	sum35
Reviewed				
		Version	Size	Page 1 Total 1
		V1.0	A4	EasyEDA.com

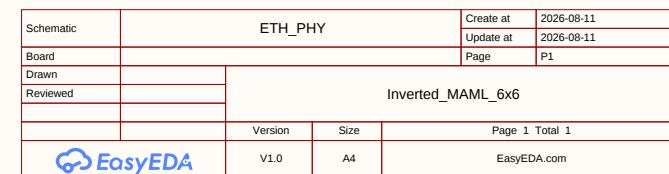




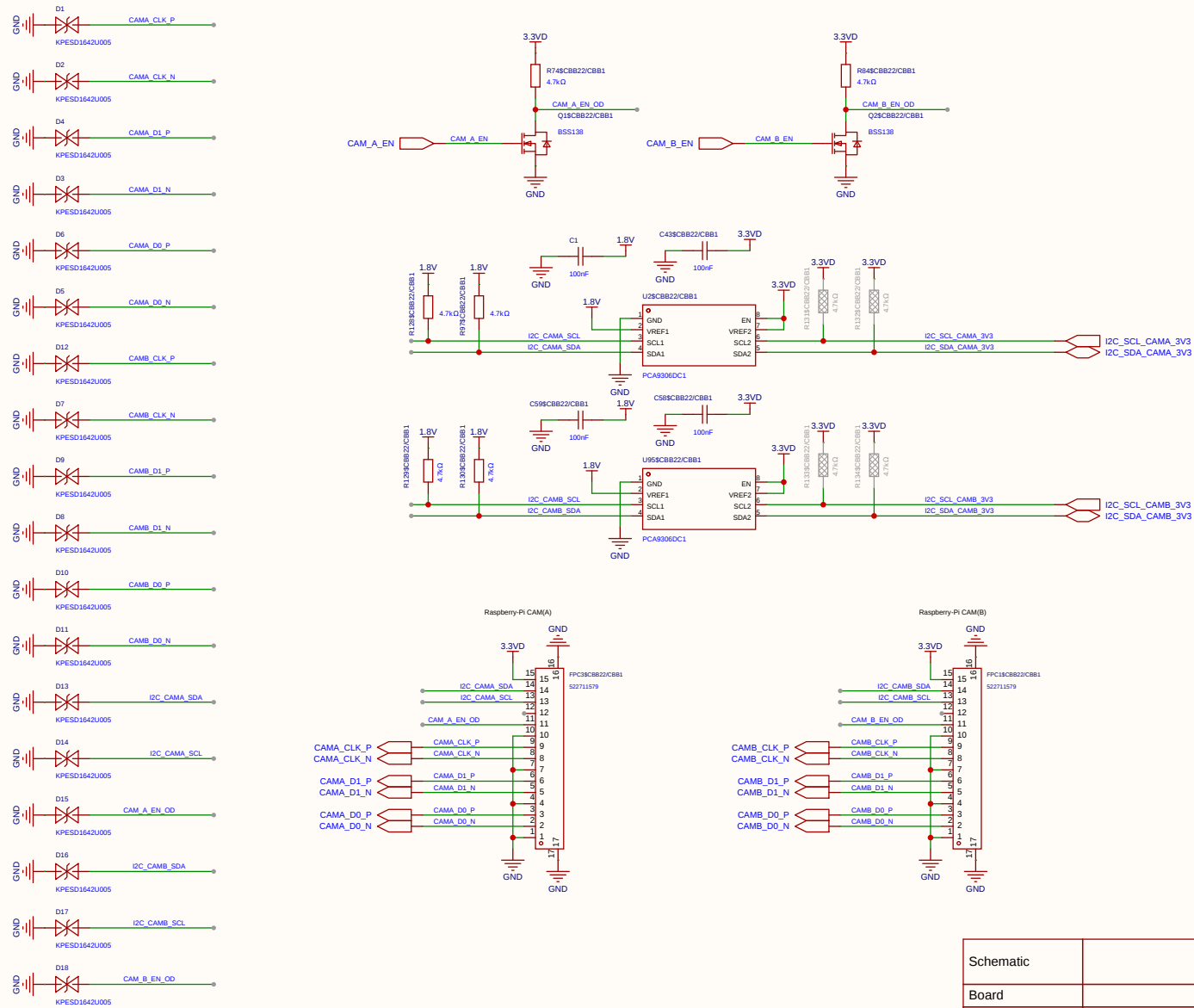
MEM0	TEST_0	SL_0	CS00	CS01	CS02
17	1	1	1	1	1
18	1	1	1	1	1
19	1	1	1	1	1

(Table 1. SP-WebServer Settings)

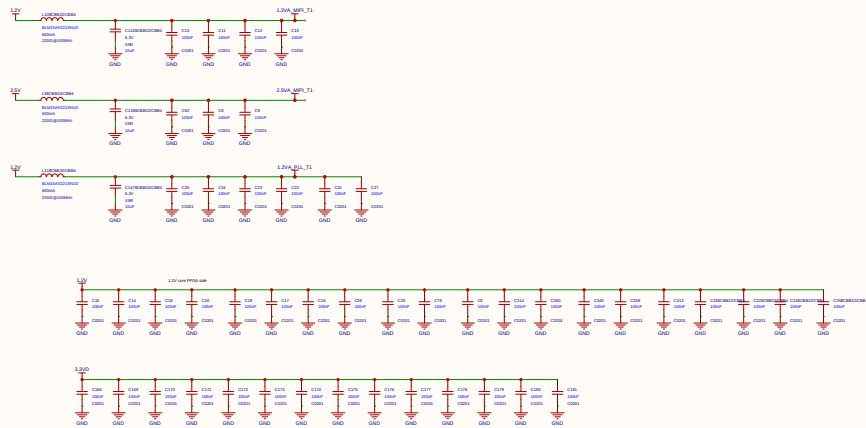


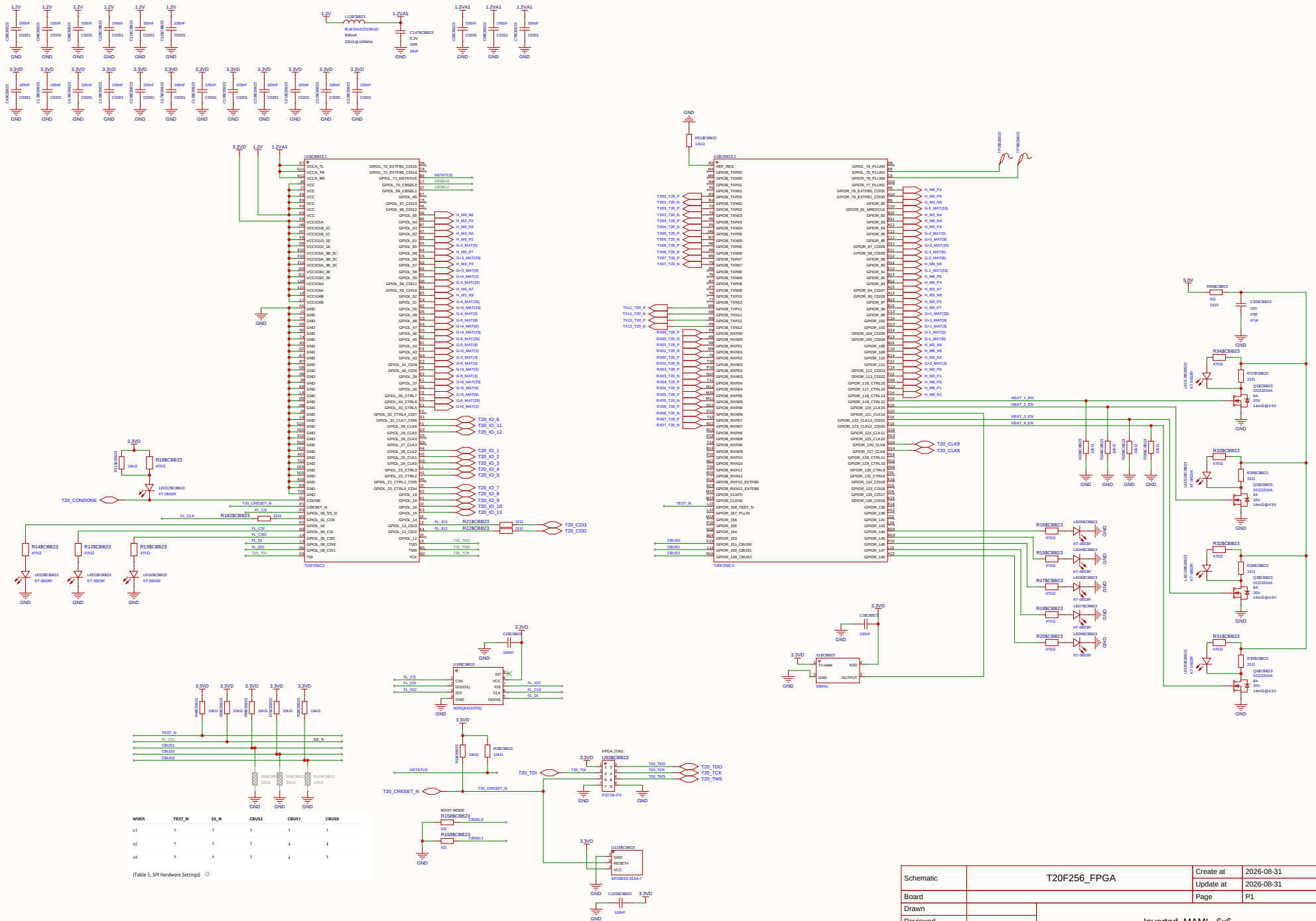


CSI- CAM is just for another project I put up this Raspberry camera Interface. (Not for Intverted MAML proof of concept)



Schematic	CAM-AB			Create at	2026-08-19
				Update at	2026-09-02
Board				Page	P1
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	





Schematic	T20F256_FPGA			Create at	2026-08-31
Board				Update at	2026-08-31
Drawn				Page	P1
Reviewed				Inverted_MAML_6x6	
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	